Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	18 June 2026
Team ID	
Project Name	Online Complaint Registration
Maximum Marks	4 Marks

Technical Architecture:

The Online Complaint Registration System follows a three-tier architecture consisting of the Presentation Layer, Application Layer, and Data Layer. The Presentation Layer is developed using React.js, providing an interactive interface for users and administrators. The Application Layer is implemented using Node.js and Express.js, which handle authentication, complaint management, status updates, and feedback processing through RESTful APIs. The Data Layer uses MongoDB to store user information, complaints, and feedback records. Communication between the frontend and backend is performed through Axios API requests, while JWT authentication ensures secure access to protected resources.

Infrastructure

- **Frontend:** React.js
- **Backend:** Node.js, Express.js
- **Database:** MongoDB
- **Authentication:** JWT
- **API Communication:** Axios
- **Version Control:** Git & GitHub

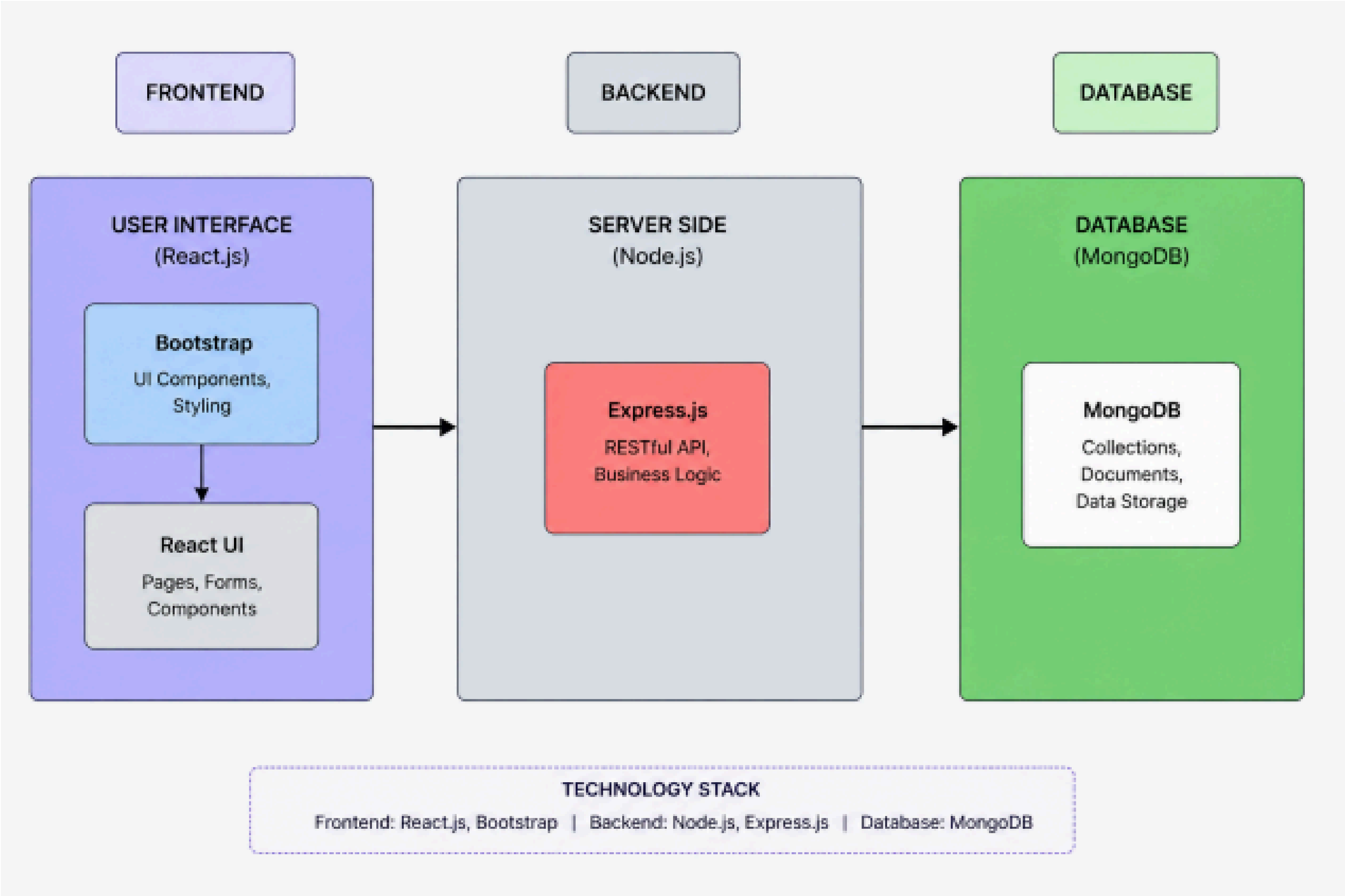

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web interface for users and admin to interact with the system	React.js, HTML, CSS, JavaScript, Bootstrap

2.	Application Logic-1	Handles authentication, complaint submission, tracking, and feedback	Node.js, Express.js
3.	Application Logic-2	Complaint status management and dashboard operations	Express.js, REST APIs
4.	Application Logic-3	Security, password hashing, token generation, and validation	JWT, bcryptjs
5.	Database	Stores user, complaint, and feedback data	MongoDB
6.	Cloud Database	Database hosting service	MongoDB Atlas
7.	File Storage	Stores project-related data and attachments	Local Storage

8. External API-1 API communication between frontend and backend Axios

9.	External API-2	Not used in current version	—
10.	Machine Learning Model	Not used in this project	—
11.	Infrastructure (Server / Cloud)	Local development and deployment environment	Node.js Server, Vercel/Render

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Full-stack development using open-source technologies	React.js, Node.js, Express.js, MongoDB
2.	Security Implementations	Secure authentication and authorization using tokens and password hashing	JWT, bcryptjs, CORS
3.	Scalable Architecture	Three-tier architecture separating frontend, backend, and database	MERN Stack Architecture

4.	Availability	Application accessible through web interface with database availability	MongoDB Atlas
5.	Performance	Fast API responses and efficient database queries	Express.js, MongoDB Indexing